

Week 2 Internship Documentation

pfSense Firewall Deployment, Suricata IDS, and pfBlockerNG Configuration

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Internship

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Objective

The objective of this lab was to deploy and configure a pfSense Firewall in a virtual environment, implement intrusion detection using Suricata, configure pfBlockerNG for GeoIP and DNS-based filtering, create administrator access rules, validate firewall functionality, and document the complete implementation process.

Lab Environment

Virtual Machines

- pfSense Firewall
 - Ubuntu Server
 - Ubuntu Desktop (GUI installed on Ubuntu Server for firewall management and testing)
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Software Used

- Oracle VirtualBox
 - pfSense CE
 - Ubuntu Server
 - Ubuntu Desktop
 - Suricata IDS/IPS
 - pfBlockerNG
 - Firefox Web Browser
 - MaxMind GeoLite2 Database
-

Virtual Network Configuration

The lab environment consisted of two virtual network adapters.

pfSense Firewall

Adapter 1 (WAN)

- Bridged Adapter
- Connected to the physical network for Internet access.

Adapter 2 (LAN)

- Internal Network
- Configured with:

LAN IP Address:
192.168.1.1/24

Ubuntu Desktop was connected to the LAN network for testing and management.

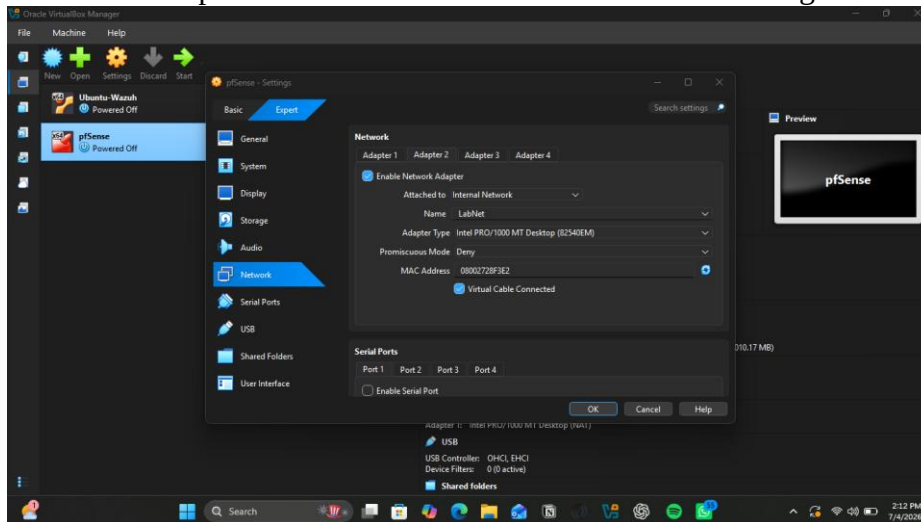


Fig1: Configuring Network Adapters
for pfSense VM

Installing pfSense

The pfSense ISO image was attached to the virtual machine and installed successfully.

Installation steps included:

- Booting from the pfSense ISO.
- Selecting Automatic ZFS partitioning.
- Completing the installation.
- Rebooting the virtual machine.

Initially, after rebooting, the installer started again because the installation ISO was still attached to the virtual machine. The installation media was removed, and the virtual machine was restarted, allowing pfSense to boot correctly.

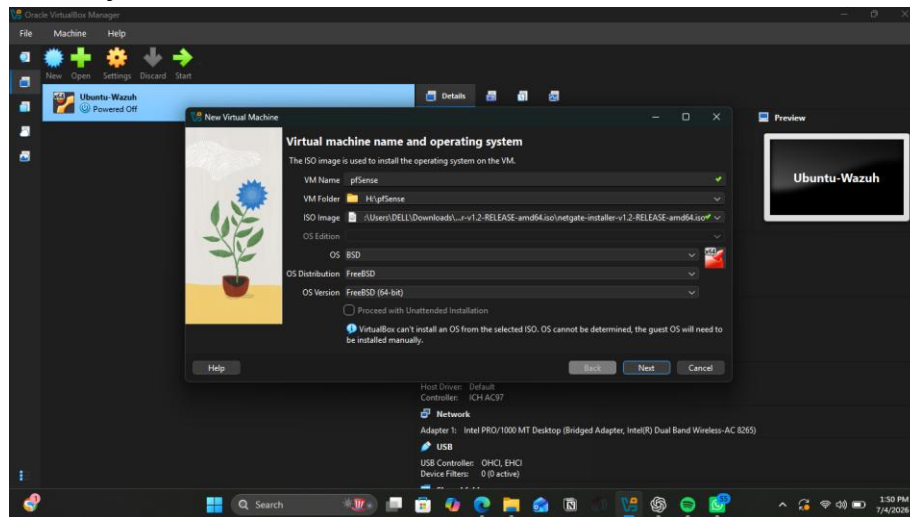


Fig 2: ISO image being attached to VM and configuring it

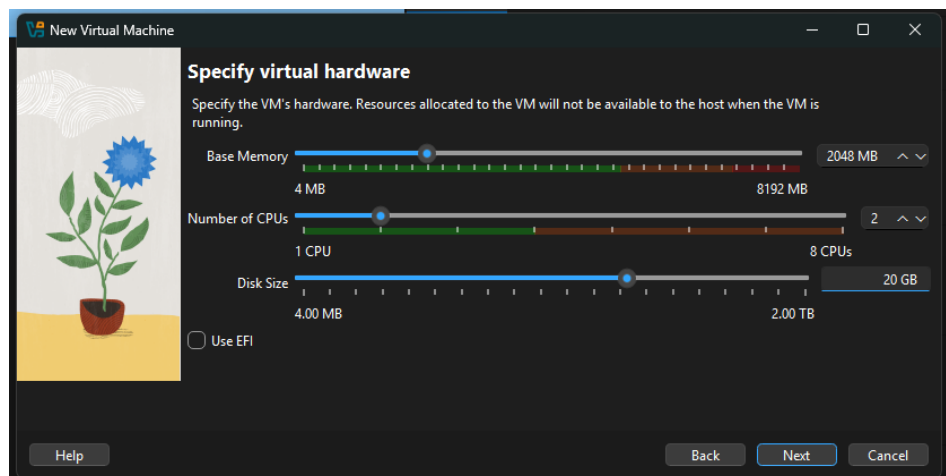


Fig3: Configuring virtual hardware for pfSense VM

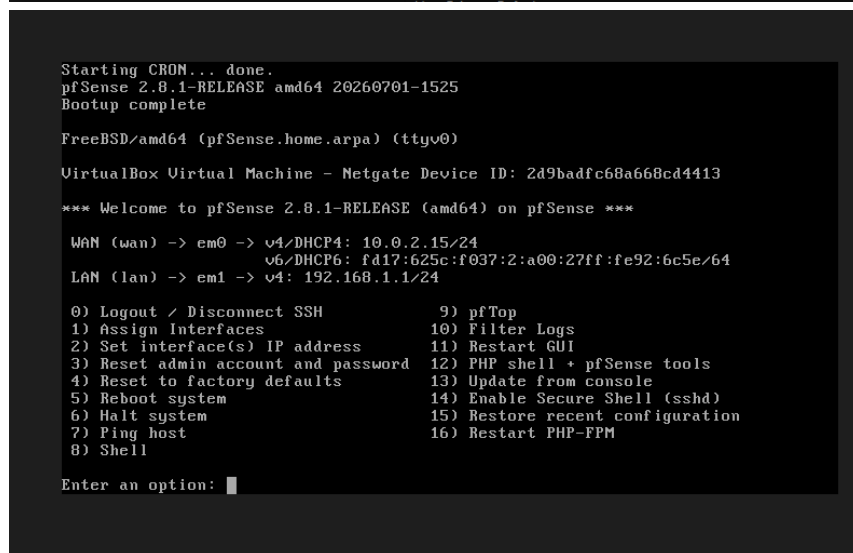


Fig4: pfSense installed succesfully

Initial pfSense Configuration

After booting successfully, the LAN interface was configured with:

```
192.168.1.1/24
```

The pfSense WebGUI was accessed through:

```
https://192.168.1.1
```

The default administrator credentials were used for the first login, after which the administrator password was changed for security.

The Setup Wizard was completed by configuring:

- Hostname
- Domain
- DNS
- Time Zone
- Administrator Password

Ubuntu Desktop Installation

Initially, Ubuntu Server was installed without a graphical desktop environment. Since pfSense management is significantly easier through a web browser, a desktop environment was installed on the existing Ubuntu Server installation.

The following commands were executed:

```
sudo apt update  
sudo apt install ubuntu-desktop
```

After installation, the virtual machine was restarted.

The existing Ubuntu Server configuration, users, installed software, and files remained unchanged while the graphical desktop environment was added successfully.

Mozilla Firefox was then used to access the pfSense WebGUI.

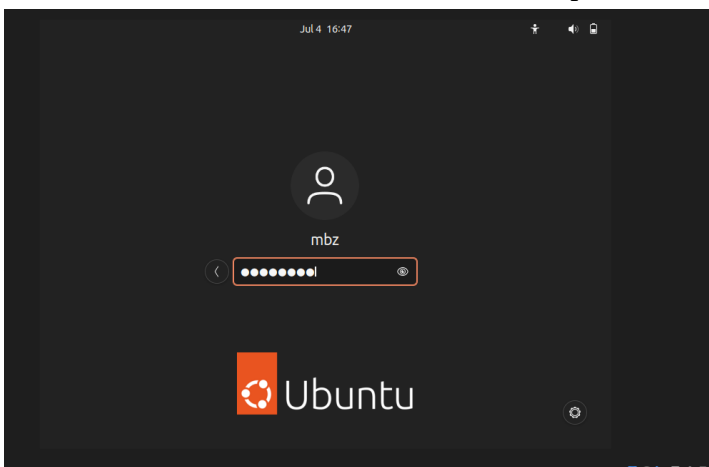


Fig5: Logging into Ubuntu Desktop after installation

Network Troubleshooting

Several networking issues were encountered during the implementation.

Initially, Ubuntu Desktop was unable to communicate with the pfSense LAN interface.

The following command was used to verify network interfaces:

```
ip -br addr
```

It was observed that the secondary interface (`enp0s8`) was in a DOWN state.

The interface was enabled using:

```
sudo ip link set enp0s8 up
```

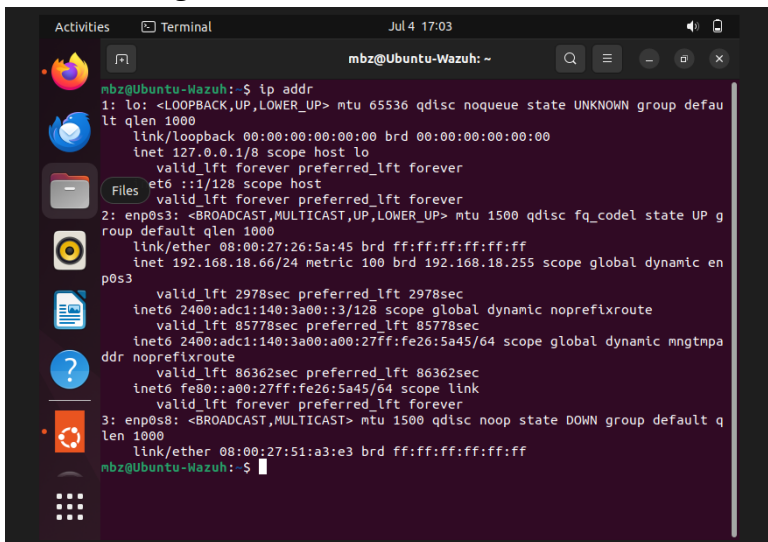
A DHCP address was requested using:

```
sudo dhclient enp0s8
```

Connectivity was verified using:

```
ping 192.168.1.1
```

After enabling the interface, communication between Ubuntu Desktop and pfSense was successfully restored.



```
mbz@Ubuntu-Wazuh:~$ ip -br addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    et6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:26:5a:45 brd ff:ff:ff:ff:ff:ff
    inet 192.168.18.66/24 metric 100 brd 192.168.18.255 scope global dynamic enp0s3
        valid_lft 2978sec preferred_lft 2978sec
    inet6 2400:adc1:140:3a00::3/128 scope global dynamic noprefixroute
        valid_lft 85778sec preferred_lft 85778sec
    inet6 2400:adc1:140:3a00:a00:27ff:fe26:5a45/64 scope global dynamic mngtmpa
        valid_lft 86362sec preferred_lft 86362sec
    inet6 fe80::a00:27ff:fe26:5a45/64 scope link
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 08:00:27:51:a3:e3 brd ff:ff:ff:ff:ff:ff
mbz@Ubuntu-Wazuh:~$
```

Fig6: Using Ubuntu Desktop's Terminal for troubleshooting network

DNS Verification

To verify that Ubuntu Desktop was using pfSense as its DNS server, the following command was executed:

```
resolvectl status
```

The output confirmed:

```
Current DNS Server:  
192.168.1.1
```

This confirmed that all DNS requests were passing through the pfSense firewall.

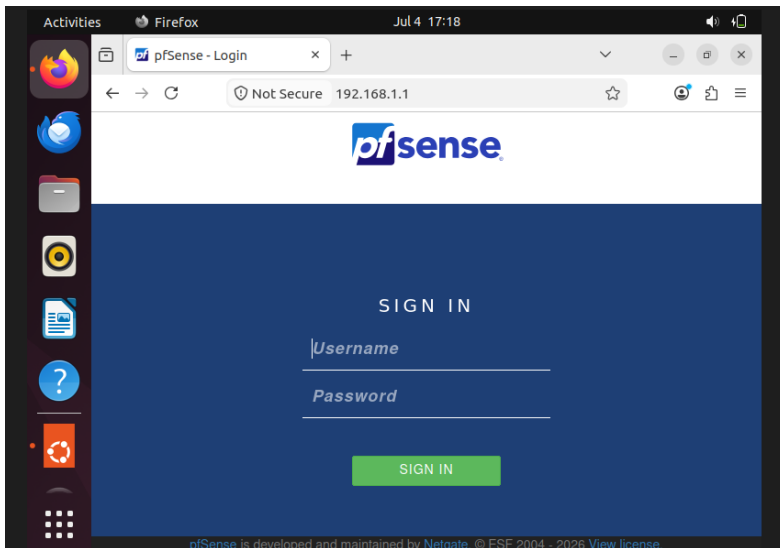


Fig7: pfSense web GUI opened

Suricata Installation

Suricata IDS/IPS was installed through:

System → Package Manager → Available Packages

Installed package:

```
Suricata 7.0.8_5
```

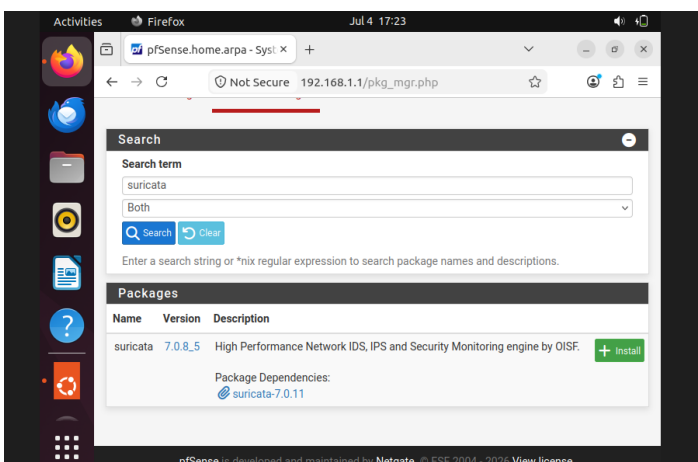


Fig8: Searching up Suricata in Package Manager to install

Suricata Configuration

Suricata was configured on the WAN interface.

The following options were enabled:

- Enable Interface
- IPS Mode
- Pattern Matcher

The Emerging Threats Open (ET Open) rule set was downloaded and installed successfully.

Traffic generated from Ubuntu Desktop produced alerts, confirming successful IDS operation.

Alerts were verified through:

Services → Suricata → Alerts

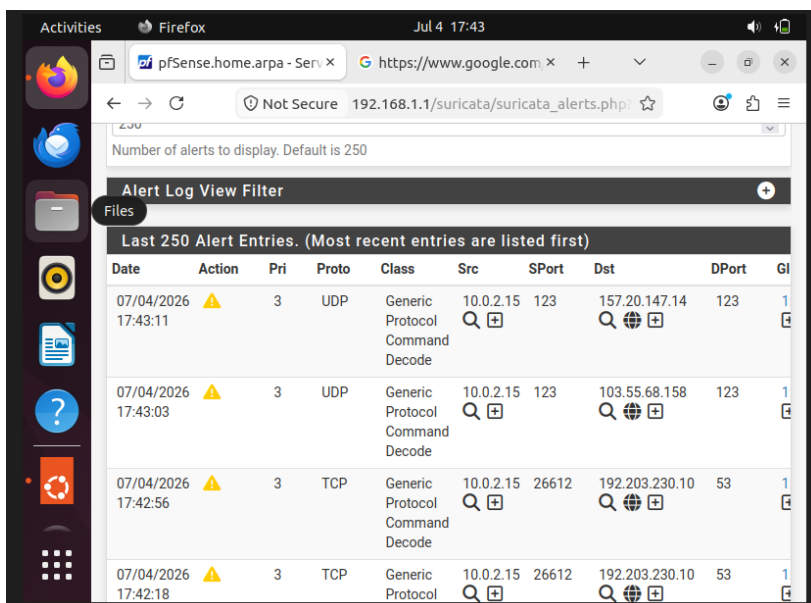


Fig9: Suricata showing alerts after being configured and tested

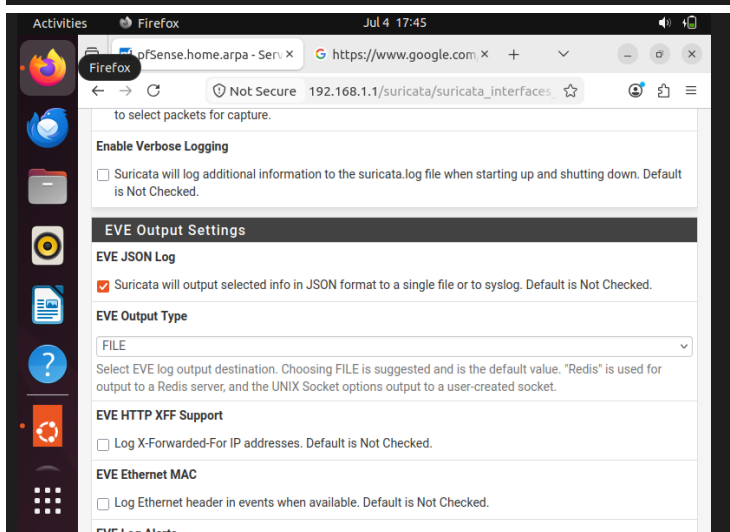


Fig10: Configuring Suricata to generate JSON log file which would be used by pfBlockerNG

Installing pfBlockerNG

The package installed was:

pfBlockerNG

The setup wizard was completed.

Configuration included:

- WAN selected as Inbound Interface
- LAN selected as Outbound Interface
- DNSBL enabled
- DNSBL Whitelist enabled

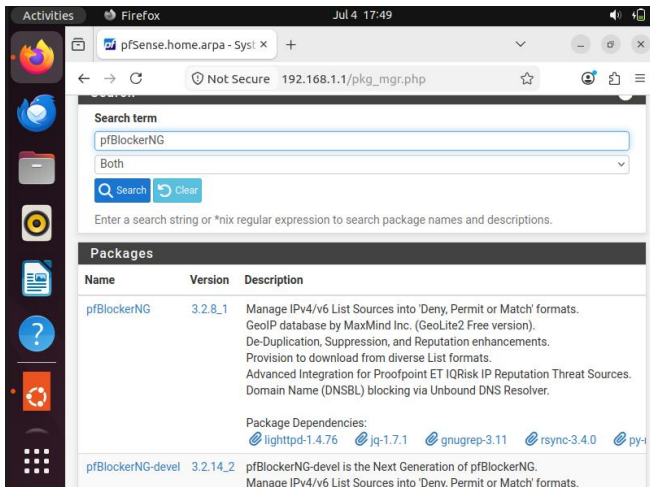


Fig11: Searching up pfBlockerNG in Package Manager to install

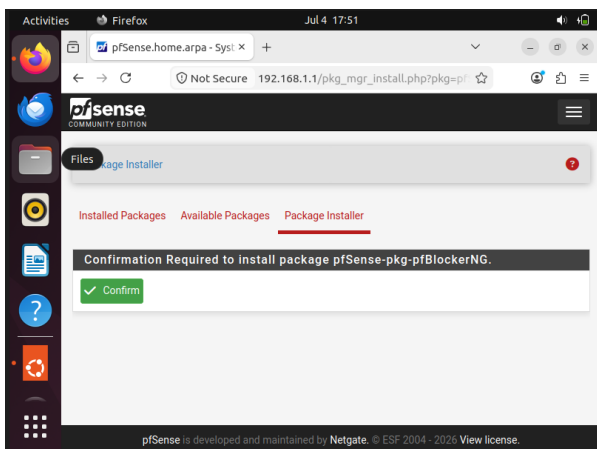
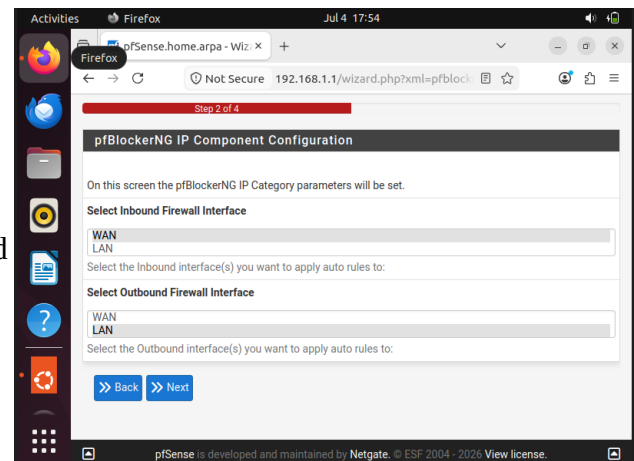


Fig12: Confirming install of pfBlockerNG

Fig 13: Selecting as Inbound Interface and LAN as Outbound Interface



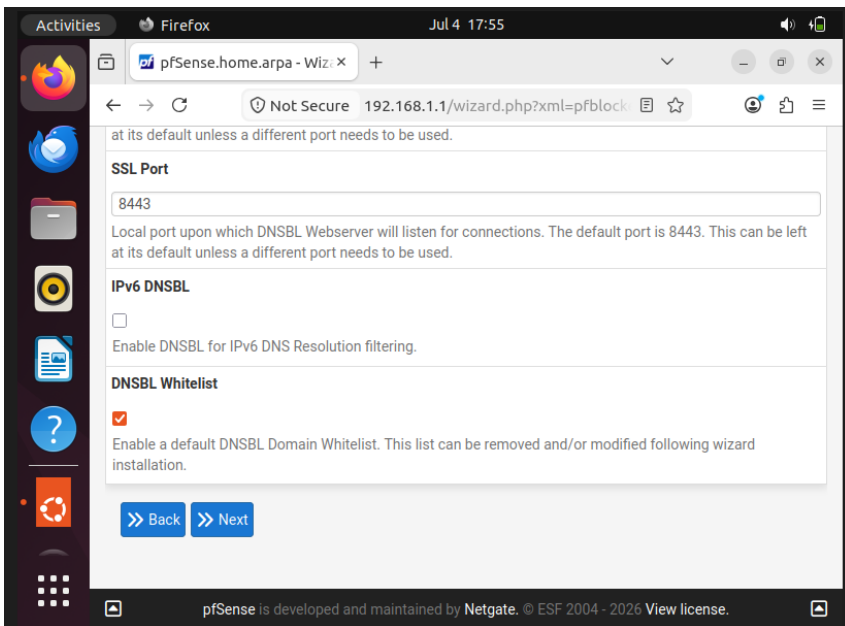


Fig14: Enabling DNSBL and DNSBL Whitelist

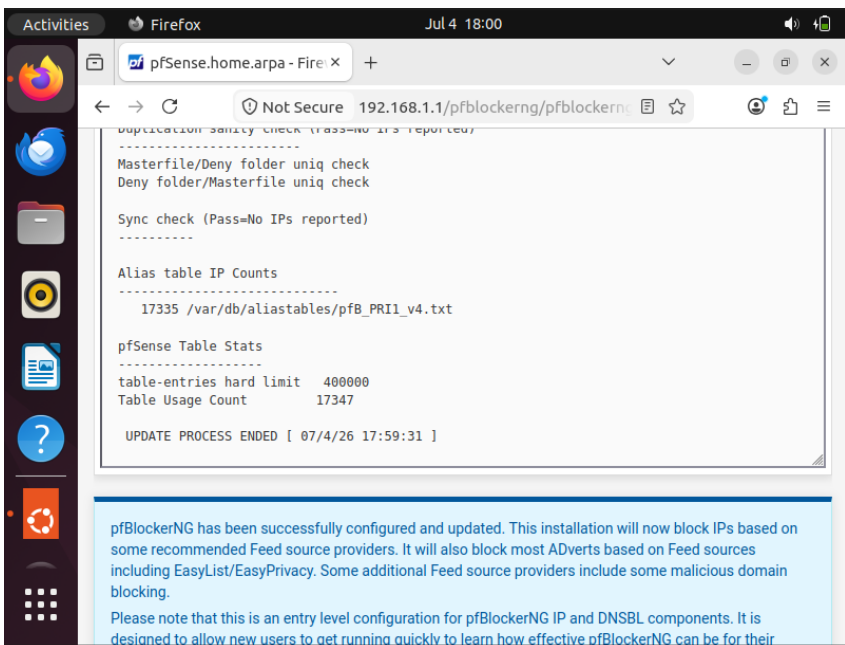


Fig15: Updated pfBlockerNG

Configuring MaxMind GeoIP

A MaxMind account was created.

A GeoLite2 License Key was generated.

Initially, downloading the GeoLite database resulted in:

401 Unauthorized

The issue was resolved by configuring the correct license key.

The GeoLite2 Country Database downloaded successfully afterwards.

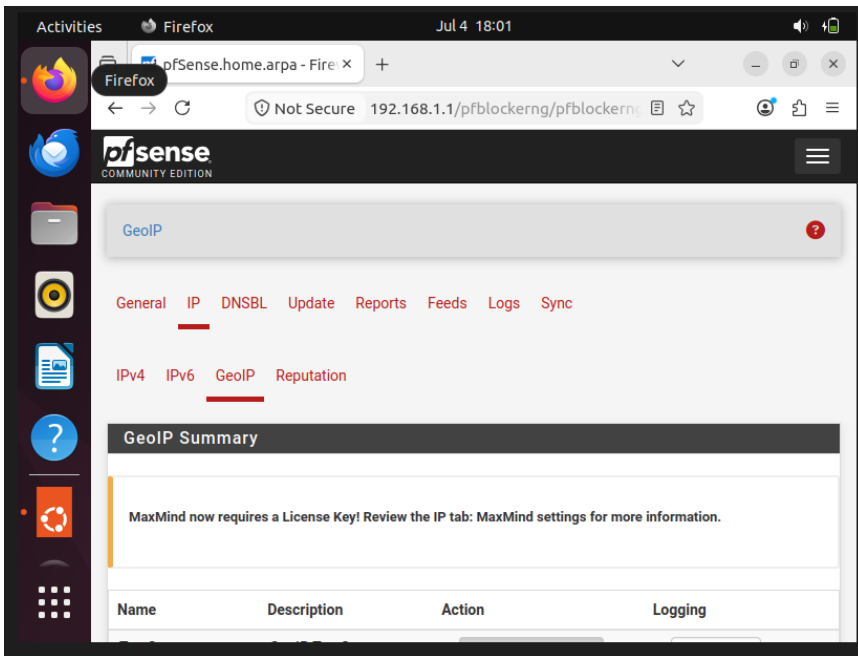


Fig16: GeoIP requiring MaxMind license key

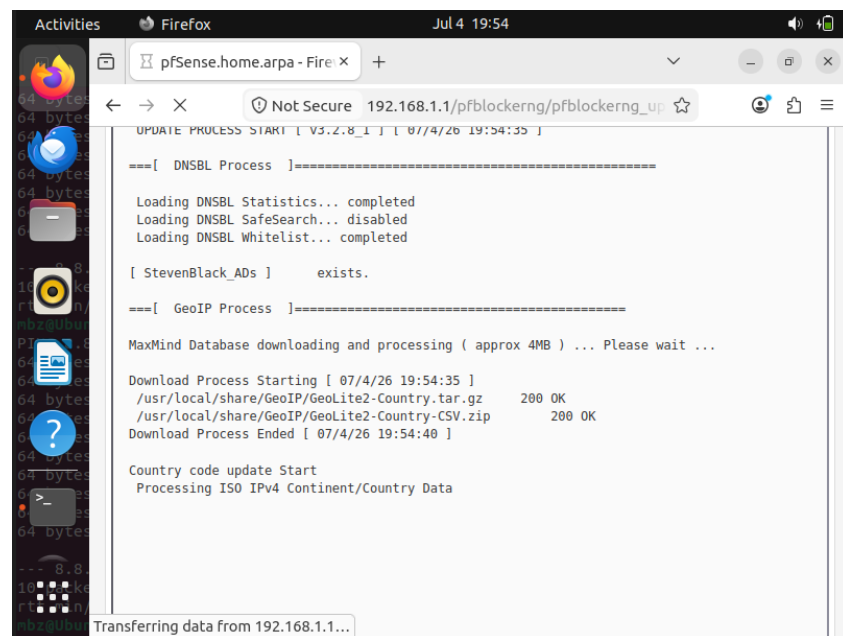


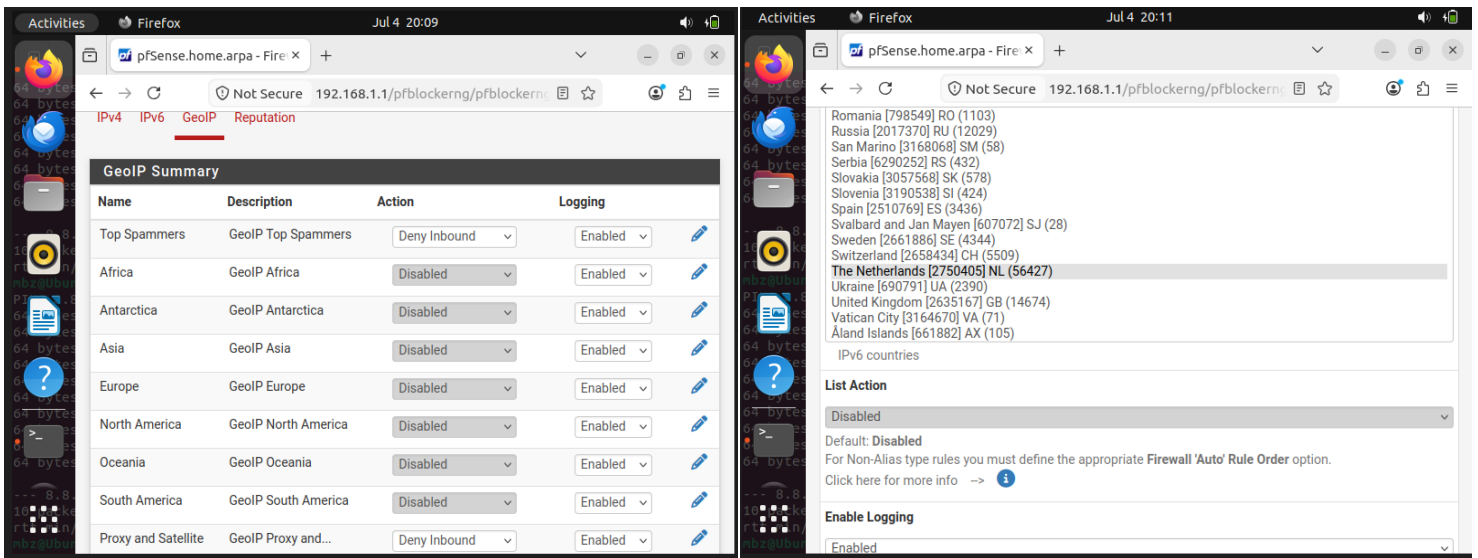
Fig17: GeoLite2 database being downloaded after entering license key and account ID in IP tab

GeoIP Configuration

GeoIP filtering was configured successfully.

Country databases were downloaded and aliases were created automatically.

The GeoIP configuration was verified successfully.



DNSBL Configuration

DNSBL was enabled within pfBlockerNG.

Initially, the following category lists were enabled:

- ShallaList
- UT1

The Social Networking category was selected within both lists.

A custom DNSBL blocklist was also created containing:

```
facebook.com
www.facebook.com
chat.com
www.chat.com
```

Initially, the custom blocklist remained disabled, causing Facebook to continue resolving normally.

After troubleshooting, the custom blocklist was enabled.

The DNS configuration was rebuilt using:

Firewall → pfBlockerNG → Update

Force Option:

Reload

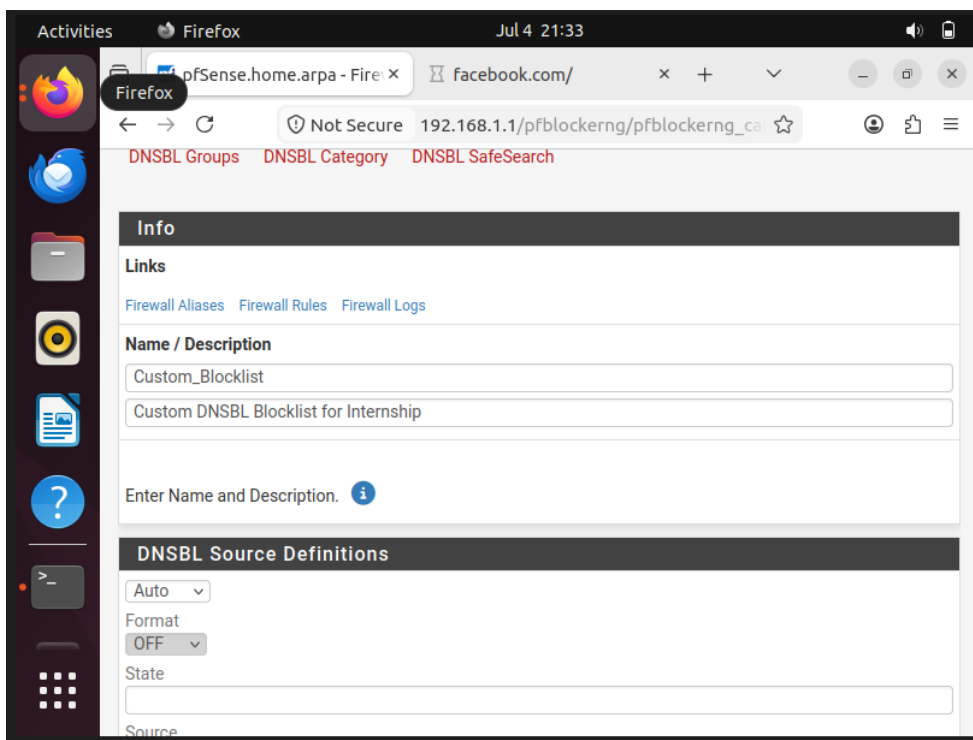
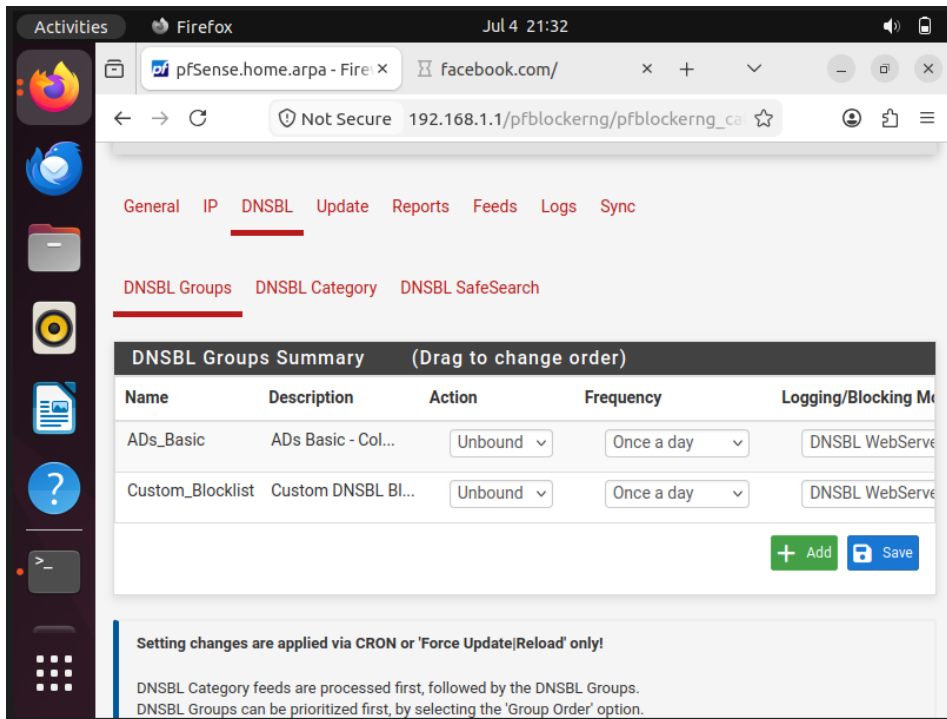
Reload Type:

DNS

Ubuntu DNS cache was then cleared using:

```
sudo resolvectl flush-caches
```

After the reload, DNSBL successfully blocked the configured websites.



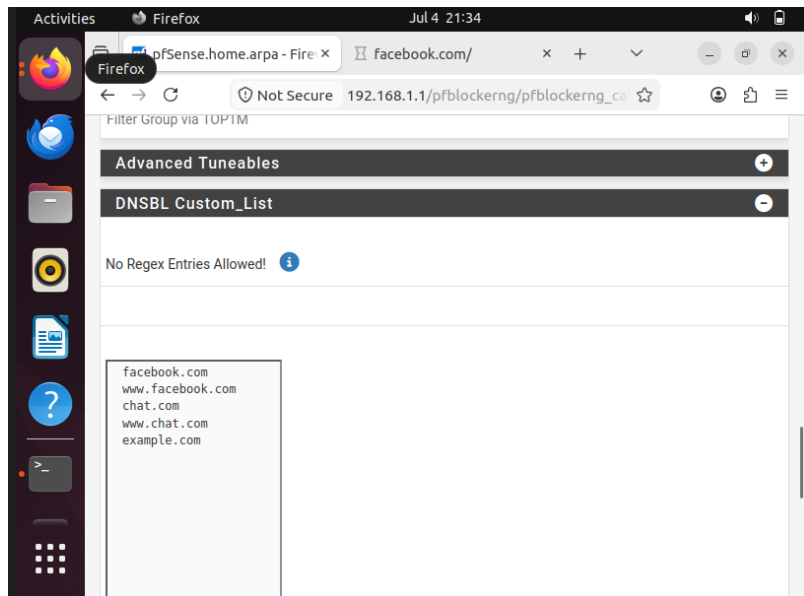


Fig 20,21 and 22: Making Custom Blocklist in DNSBL Groups to block facebook.com, chat.com and example.com

Administrator WebGUI Access Rule

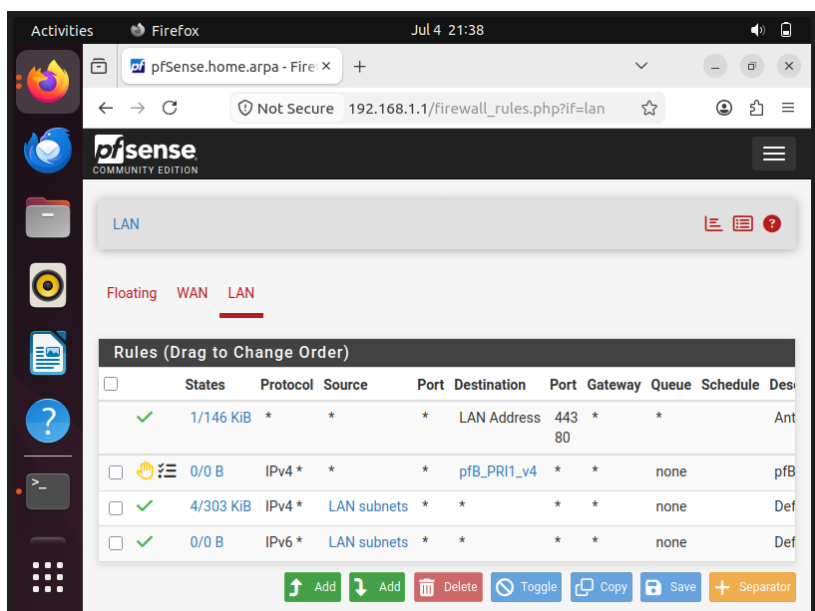
A firewall rule was created to allow secure administrator access to the pfSense WebGUI.

Configuration:

- Action: Pass
- Interface: LAN
- Protocol: TCP
- Source: LAN Subnets
- Destination: This Firewall
- Destination Port: HTTPS (443)

Description:

Allow Admin WebGUI Access



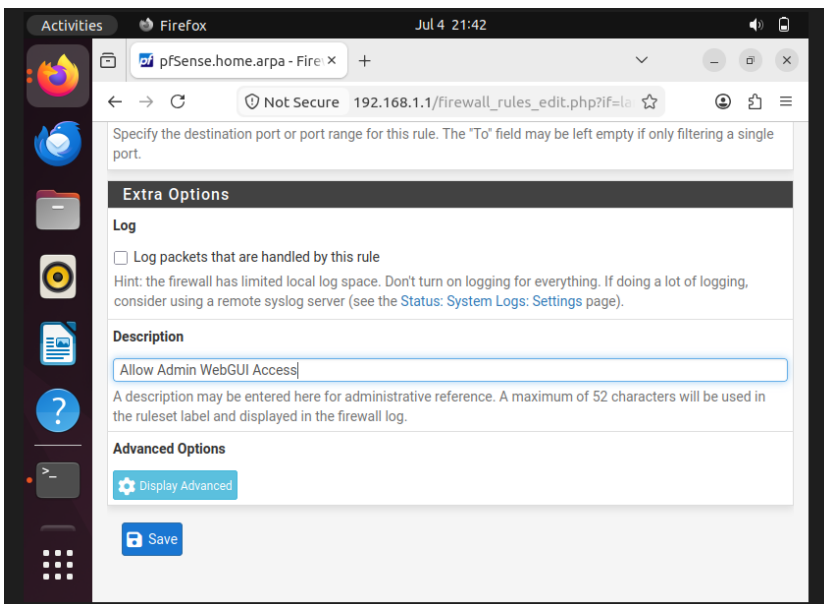
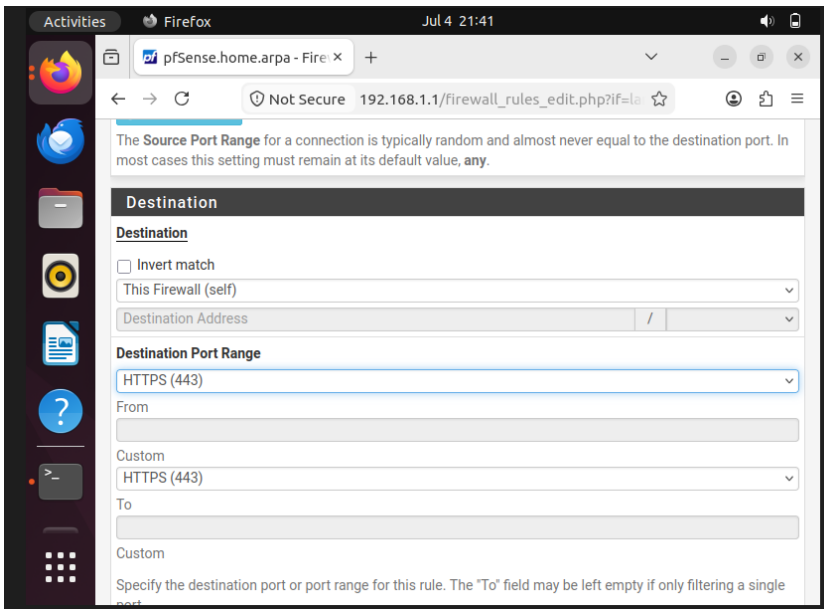
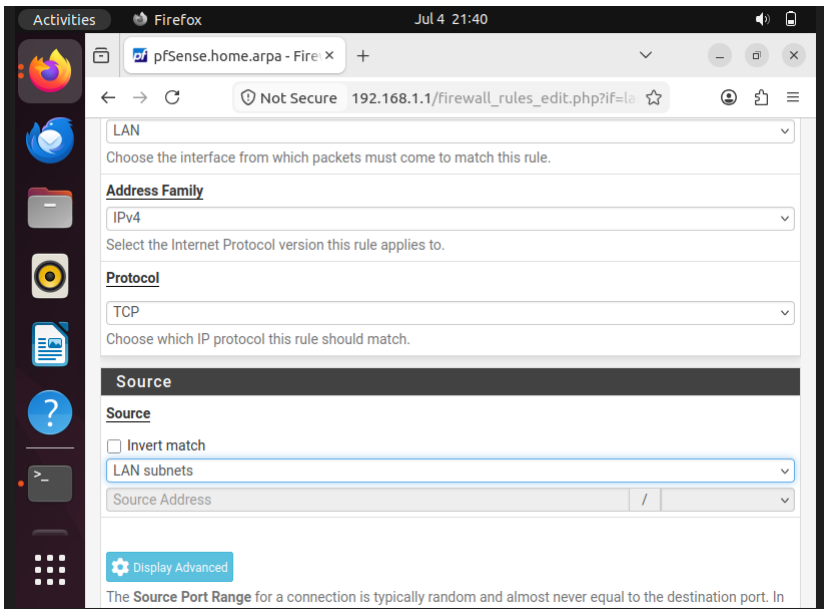


Fig 23,24,25 and 26: In Firewall → Rules → LAN, added another Rule and saved

Validation Testing

The firewall implementation was validated through multiple tests.

DNS Testing

The following command was used:

```
nslookup facebook.com
```

Initially, Facebook resolved successfully.

After enabling the custom DNSBL blocklist and performing a DNS reload, Facebook access was blocked successfully.

Website Blocking

The following websites were tested:

- facebook.com
- www.facebook.com
- chat.com
- www.chat.com

Both Facebook and Chat.com were successfully blocked through DNSBL.

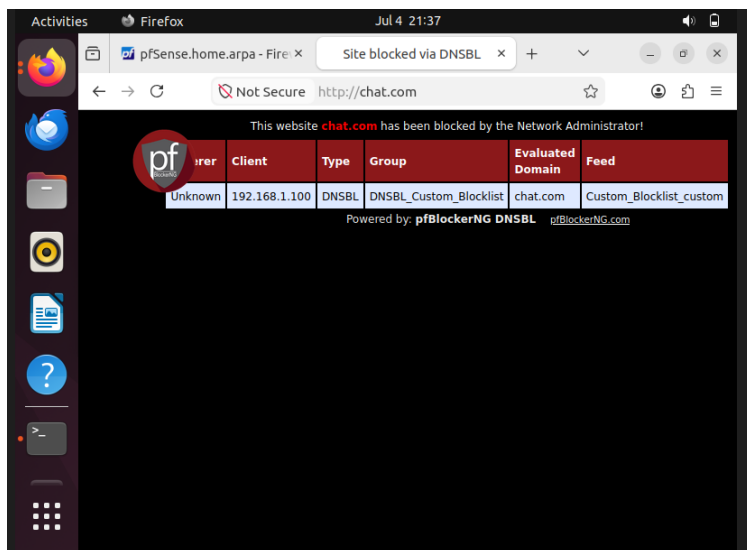


Fig 27: pfBlockerNG blocking chat.com

Suricata Testing

Network traffic generated alerts inside Suricata.

Alerts were successfully verified.

GeoIP Verification

The GeoLite2 database downloaded successfully.

GeoIP aliases were created successfully.

Commands Used During the Lab

Check interfaces:

```
ip -br addr
```

Enable LAN interface:

```
sudo ip link set enp0s8 up
```

Request DHCP:

```
sudo dhclient enp0s8
```

Verify DNS server:

```
resolvectl status
```

Flush DNS cache:

```
sudo resolvectl flush-caches
```

DNS lookup:

```
nslookup facebook.com
```

Ping pfSense:

```
ping 192.168.1.1
```

Issues Encountered and Resolutions

Issue 1

After installation, pfSense repeatedly booted into the installer.

Resolution

The installation ISO was removed from the virtual machine before rebooting.

Issue 2

Ubuntu Desktop could not communicate with pfSense.

Resolution

The LAN interface (`enp0s8`) was down and was enabled manually before requesting a DHCP address.

Issue 3

GeoLite2 download failed with a 401 Unauthorized error.

Resolution

A valid MaxMind License Key was created and configured.

Issue 4

DNSBL failed to block Facebook.

Resolution

DNS troubleshooting confirmed Ubuntu was using pfSense as its DNS server. The issue was traced to the custom DNSBL blocklist being disabled. After enabling the blocklist and reloading DNS, website blocking functioned correctly.

Results

The following objectives were achieved successfully:

- Successfully deployed pfSense Firewall.
 - Configured WAN and LAN interfaces.
 - Installed Ubuntu Desktop for GUI-based administration.
 - Installed and configured Suricata IDS/IPS.
 - Installed and configured pfBlockerNG
 - Configured GeoIP country filtering.
 - Configured DNSBL filtering.
 - Created a custom DNSBL blocklist.
 - Successfully blocked Facebook and Chat.com.
 - Created administrator WebGUI access rules.
 - Validated firewall events using Suricata and pfBlockerNG.
 - Verified DNS filtering and firewall functionality.
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Conclusion

This lab provided practical experience in deploying and configuring an enterprise-grade firewall using pfSense. Network segmentation, intrusion detection, GeoIP filtering, DNS-based website blocking, and secure administrator access were successfully implemented. Throughout the implementation, multiple troubleshooting scenarios—including network interface failures, DNS configuration issues, GeoLite authentication errors, and DNSBL configuration problems—were identified and resolved. The completed environment demonstrates a layered security architecture suitable for Blue Team and SOC operations and fulfills all Week 2 internship requirements.